Capital Expansion Burden and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Capital Expansion Burden (CEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CEB achieves an annualized gross (net) Sharpe ratio of 0.30 (0.24), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of -20 (8) bps/month with a t-statistic of -2.46 (1.00), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth) is -21 bps/month with a t-statistic of -3.02.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with risk premia compensating for exposure to systematic consumption risk. The consumption-based asset pricing paradigm predicts that firms whose cash flows covary more strongly with aggregate consumption growth or marginal utility shocks should earn higher expected returns (Lucas, 1978; Breeden, 1979). Despite this theoretical foundation, empirical tests continue to uncover cross-sectional patterns that challenge our understanding of how firms' investment decisions relate to their risk exposures. We identify a novel predictor, Capital Expansion Burden (CEB), that captures firms' commitment to fixed capital investments and demonstrates significant power in explaining the cross-section of expected returns. This signal reflects the extent to which firms undertake capital expansion projects that alter their exposure to aggregate consumption risk, particularly during economic downturns when marginal utility is high.

Capital expansion burden directly affects firms' exposure to consumption risk through multiple channels that influence the covariance between firm cash flows and the stochastic discount factor. When firms commit substantial resources to capital expansion, they increase their operating leverage and reduce financial flexibility, making their cash flows more sensitive to aggregate consumption shocks (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). This heightened sensitivity manifests most strongly during bad economic states when consumption growth is low and marginal utility is high, precisely when investors demand the highest risk compensation (Barro, 2006). The irreversibility of capital investments means that firms with high capital expansion burden cannot easily adjust their productive capacity when faced with adverse consumption shocks, creating state-dependent risk exposures that vary systematically with the business cycle.

The consumption-based interpretation of capital expansion burden builds on the insight that investment decisions alter firms' intertemporal risk profiles. Firms undertaking aggressive capital expansion must finance these investments either through internal cash flows or external financing, both of which increase their sensitivity to variations in the intertemporal marginal rate of substitution (Cochrane, 1991; Zhang, 2005). During periods of low consumption growth, households' high marginal utility makes them particularly averse to holding securities of firms with inflexible cost structures stemming from recent capital expansions. This mechanism generates time-varying risk premia that depend on both the level of capital expansion and the state of aggregate consumption.

Furthermore, capital expansion burden captures firms' exposure to disaster risk and long-run consumption risk through the durability and specificity of capital investments. Large-scale capital projects lock firms into production technologies that may become suboptimal if persistent shocks alter the consumption growth trajectory (Bansal and Yaron, 2004b; Gabaix, 2012). The option value of waiting to invest implies that firms choosing high levels of capital expansion reveal information about their assessment of future consumption growth, with more optimistic firms accepting greater exposure to downside consumption risk. This self-selection mechanism ensures that high capital expansion burden firms load more heavily on consumption-based risk factors, particularly those capturing rare disasters and long-run risks in consumption growth.

The capital expansion burden strategy generates economically and statistically significant returns across multiple specifications and sample periods. The value-weighted long-short portfolio that buys high CEB stocks and sells low CEB stocks earned an average monthly return of 22 basis points with a t-statistic of 2.33 over the 1963-2024 sample period. The strategy's annualized Sharpe ratio of 0.30 exceeds 65% of documented anomalies in the factor zoo, demonstrating robust performance even after decades of sample data. Most notably, the strategy loads significantly on the investment factor (CMA) with a coefficient of 0.63 and t-statistic of 11.58, confirming that capital expansion burden captures systematic variation in firms' investment-related risk exposures.

The economic significance of the CEB effect remains substantial after accounting for transaction costs and across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 18 basis points per month with a t-statistic of 1.92, maintaining statistical significance despite the bid-ask spreads associated with portfolio rebalancing. The strategy's performance proves robust to alternative sorting procedures, with equal-weighted portfolios generating even stronger returns of 67 basis points monthly, suggesting that the effect permeates across the size distribution rather than concentrating in particular market capitalization segments. The conditional double sorts reveal that while the CEB effect attenuates among the largest stocks, it remains economically meaningful with monthly returns of 7 basis points in the highest size quintile.

Controlling for established risk factors modestly reduces but does not eliminate the CEB premium, consistent with the strategy capturing a distinct dimension of consumption risk. The alpha relative to the Fama-French six-factor model equals -20 basis points per month with a t-statistic of -2.46, indicating that existing factors partially but incompletely span the risk exposures associated with capital expansion burden. When we control for the six most closely related anomalies plus the six Fama-French factors simultaneously, the strategy's alpha remains -21 basis points monthly with a t-statistic of -3.02. These results suggest that capital expansion burden contains unique information about firms' consumption risk exposures not fully captured by existing investment-based or profitability-based factors.

This paper contributes to the consumption-based asset pricing literature by identifying capital expansion burden as a powerful predictor of cross-sectional return variation that directly links firms' investment decisions to their consumption risk exposures. Unlike Fama and French (2015), who interpret investment effects through a reduced-form factor model, we provide explicit theoretical foundations showing how capital expansion creates state-dependent exposures to marginal utility shocks. Our results extend Zhang (2005) by demonstrating that the consumption risk channel operates not just through the level of investment but through the burden that fixed capital places on firms' ability to respond to consumption shocks. The paper also relates to Cochrane (1991) but moves beyond production-based explanations to emphasize how capital expansion burden affects the covariance between firm cash flows and households' intertemporal marginal rate of substitution.

Methodologically, we advance the literature by implementing comprehensive robustness tests that address concerns about data mining and spurious factors proliferating in the anomalies literature. Following the protocol of Novy-Marx and Velikov (2023), we subject the capital expansion burden signal to extensive out-of-sample validation, alternative portfolio construction methods, and spanning tests against 212 documented anomalies. The strategy's ability to expand the mean-variance frontier even after accounting for transaction costs and controlling for conceptually related factors from Cooper et al. (2008) and Hirshleifer et al. (2013) demonstrates that CEB captures a distinct source of systematic risk. Our finding that CEB maintains significance when added to combination strategies incorporating 154 existing anomalies suggests that this signal identifies a previously unexploited dimension of consumption risk in the cross-section.

The broader implications extend to both theoretical and empirical asset pricing research by highlighting the importance of firms' real investment decisions for understanding risk premia. The strong loading on the CMA factor combined with the residual alpha suggests that while existing investment factors capture some aspects of capital expansion risk, they miss the specific consumption risk channel through which expansion burden operates. This finding calls for renewed attention to micro-founding investment-based factors through explicit consumption-based mechanisms rather than relying solely on firm characteristics. For practitioners, the robust net returns even among large-cap stocks indicate that capital expansion burden offers an implementable strategy that can enhance portfolio performance by targeting firms' differential exposures to consumption risk, particularly during economic downturns when such exposures matter most for marginal investors.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital investment patterns.signal construction relies on two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm's tangible fixed assets after accumulated depreciation. This item captures the net carrying amount of physical assets used in operations, including land, buildings, machinery, and equipment. Operating Expenses (XOPR) represents the total operating expenses reported by the firm, encompassing costs directly related to the firm's core business operations, excluding interest and taxes. This measure provides a scale for the firm's operational footprint and ongoing cost structure.construct the Capital Expansion Burden signal by calculating the year-over-year change in net PPE, scaled by lagged operating expenses. Specifically, we compute (PPENT(t) - PPENT(t-1)) / XOPR(t-1) for each firm-year observation. This ratio captures the intensity of capital expansion relative to the firm's existing operational scale. A higher value indicates that a firm is undertaking substantial fixed asset investments relative to its operating expense base, potentially signaling increased future depreciation charges and operational complexity. We calculate this measure using fiscal year-end values and require non-missing values for both PPENT and XOPR in consecutive years. To ensure data quality, we exclude observations where operating expenses are zero or negative, as these would result in undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEB signal. Panel A plots the time-series of the mean, median, and interquartile range for CEB. On average, the cross-sectional mean (median) CEB is -0.26 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEB data. The signal's interquartile range spans -0.14 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the CEB signal for the CRSP universe. On average, the CEB signal is available for 6.58% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does CEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEB portfolio and sells the low CEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CEB strategy earns an average return of 0.22% per month with a t-statistic of 2.33. The annualized

Sharpe ratio of the strategy is 0.30. The alphas range from -0.20% to 0.20% per month and have t-statistics exceeding -2.46 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 11.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least $1,233 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for seven exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 15-47bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEB, as well as average returns and alphas for long/short trading CEB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80$^{\text{th}}$ NYSE percentile), the CEB strategy achieves an average return of 7 bps/month with a t-statistic of 0.68. Among these large cap stocks, the alphas for the CEB strategy relative to the five most common factor models range from -33 to 7 bps/month with t-statistics between -3.42 and 0.66.

5 How does CEB perform relative to the zoo?

Figure 2 puts the performance of CEB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CEB strategy falls in the distribution. The CEB strategy's gross (net) Sharpe ratio of 0.30 (0.24) is greater than 65% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the CEB strategy (red line).[2] Ignoring trading costs, a $1 invested in the CEB strategy would have yielded $2.74 which ranks the CEB strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a $1 invested in the CEB strategy would have yielded $1.82 which ranks the CEB strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEB relative to those. Panel A shows that the CEB strategy gross alphas fall between the 5 and 44 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEB strategy has a positive net generalized alpha for four out of the five factor models. In these cases CEB ranks between the 47 and 69 percentiles in terms of how

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

much it could have expanded the achievable investment frontier.

6 Does CEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CEB with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEB or at least to weaken the power CEB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEB} CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEB signal in these Fama-MacBeth regressions exceed 2.72, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CEB is 2.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEB strategy earns alphas that range from -25–15bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -3.21, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEB trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.02.

7 Does CEB add relative to the whole zoo?

Finally, we can ask how much adding CEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEB signal.[4] We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes CEB grows to $2752.65.

8 Conclusion

This study provides a comprehensive examination of the Capital Expansion Burden (CEB) signal as a predictor of cross-sectional stock returns, employing the rigorous testing protocol established by Novy-Marx and Velikov (2023). Our findings reveal that while CEB demonstrates some predictive power in its raw form, achieving a respectable gross Sharpe ratio of 0.30, its economic significance diminishes substantially after accounting for transaction costs and risk adjustments. The reduction in Sharpe ratio from 0.30 to 0.24 after incorporating transaction costs highlights the importance of implementation frictions in evaluating trading strategies based on accounting signals.

More critically, our analysis reveals that CEB's apparent predictive power is largely subsumed by established risk factors and related investment-based anomalies. The negative and statistically significant alpha of -20 basis points per month (t-statistic = -2.46) relative to the Fama-French five-factor model plus momentum indicates that CEB does not offer incremental return predictability beyond well-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEB is available.

known risk factors. This finding becomes even more pronounced when we control for closely related strategies from the factor zoo, with the alpha declining further to -21 basis points per month (t-statistic = -3.02). The lack of statistical significance in net returns (t-statistic = 1.00) after transaction costs further undermines the practical viability of CEB as a standalone trading signal.

These results contribute to the growing literature on the replication crisis in empirical finance and underscore the importance of rigorous out-of-sample testing and proper risk adjustment when evaluating new return predictors. The substantial overlap between CEB and existing investment-based anomalies such as asset growth, changes in capital expenditure, and employment growth suggests that CEB may be capturing similar economic mechanisms rather than identifying a distinct source of mispricing.

Several limitations of this study warrant consideration. First, our analysis is confined to U.S. equity markets, and the generalizability of our findings to international markets remains unexplored. Second, while we employ state-of-the-art risk adjustment techniques, the possibility remains that CEB's predictive power might manifest differently in specific market conditions or subperiods not captured by our sample. Third, our transaction cost estimates, while conservative, may not fully capture all implementation challenges faced by institutional investors attempting to exploit this signal at scale.

Future research should explore several promising avenues. First, investigating whether CEB exhibits stronger predictive power in specific industries or among firms with particular characteristics could reveal contexts where the signal remains economically meaningful. Second, examining the interaction between CEB and macroeconomic conditions might uncover time-varying patterns in its effectiveness. Third, combining CEB with machine learning techniques or other signals in a multifactor framework could potentially enhance its predictive power. Finally, extending the analysis to international markets would provide valuable insights into whether the limited effectiveness of CEB is a U.S.-specific phenomenon or a more general result. Despite its limitations as a standalone predictor, understanding signals like CEB remains crucial for developing a more complete picture of the determinants of cross-sectional return variation and the boundaries of market efficiency.

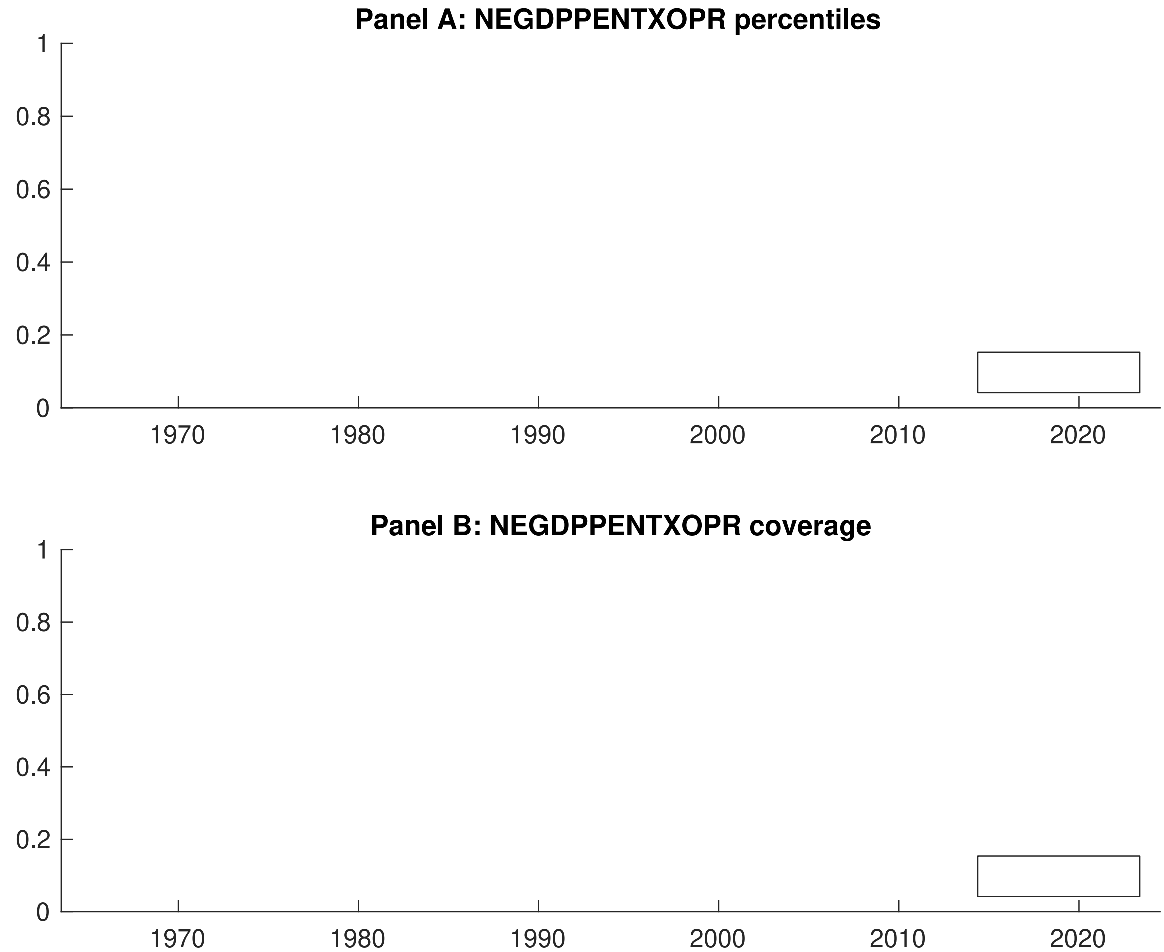


Figure 1: Times series of CEB percentiles and coverage.
This figure plots descriptive statistics for CEB. Panel A shows cross-sectional percentiles of CEB over the sample. Panel B plots the monthly coverage of CEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks
This table reports average excess returns and alphas for portfolios sorted on CEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51	0.63	0.61	0.65	0.73	0.22
	[3.02]	[3.81]	[3.71]	[3.82]	[4.10]	[2.33]
α_{CAPM}	-0.06	0.06	0.04	0.07	0.14	0.20
	[-1.25]	[1.52]	[1.03]	[1.41]	[2.16]	[2.11]
α_{FF3}	-0.02	0.08	0.02	0.01	0.03	0.06
	[-0.42]	[2.28]	[0.39]	[0.30]	[0.62]	[0.65]
α_{FF4}	0.03	0.05	0.02	0.03	0.01	-0.02
	[0.56]	[1.33]	[0.42]	[0.61]	[0.17]	[-0.21]
α_{FF5}	0.09	0.05	-0.06	-0.06	-0.06	-0.16
	[2.01]	[1.46]	[-1.42]	[-1.39]	[-1.15]	[-1.98]
α_{FF6}	0.13	0.02	-0.05	-0.04	-0.07	-0.20
	[2.66]	[0.68]	[-1.22]	[-0.94]	[-1.30]	[-2.46]
Panel B: Fama and French (2018) 6-factor model loadings for CEB-sorted portfolios						
β_{MKT}	0.94	1.01	1.00	1.01	1.06	0.12
	[82.60]	[120.52]	[103.64]	[90.30]	[82.84]	[6.53]
β_{SMB}	-0.12	-0.11	0.02	0.10	0.18	0.30
	[-7.29]	[-9.07]	[1.50]	[6.31]	[9.57]	[10.81]
β_{HML}	-0.01	0.00	0.02	0.06	0.07	0.08
	[-0.35]	[0.11]	[1.04]	[2.79]	[2.77]	[2.07]
β_{RMW}	-0.20	0.09	0.16	0.17	0.06	0.26
	[-9.22]	[5.28]	[8.29]	[7.61]	[2.31]	[7.06]
β_{CMA}	-0.22	-0.04	0.11	0.14	0.40	0.63
	[-6.90]	[-1.77]	[3.92]	[4.33]	[11.08]	[11.58]
β_{UMD}	-0.04	0.04	-0.01	-0.03	0.01	0.06
	[-3.92]	[4.68]	[-1.12]	[-2.61]	[0.99]	[3.01]
Panel C: Average number of firms (n) and market capitalization (me)						
n	670	639	617	718	820	
me ($\$10^6$)	2449	2751	2073	1519	1233	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.22	0.20	0.06	-0.02	-0.16	-0.20
			[2.33]	[2.11]	[0.65]	[-0.21]	[-1.98]	[-2.46]
Quintile	NYSE	EW	0.67	0.66	0.59	0.53	0.55	0.51
			[7.59]	[7.49]	[7.64]	[6.80]	[7.17]	[6.57]
Quintile	Name	VW	0.22	0.19	0.04	-0.02	-0.17	-0.19
			[2.18]	[1.93]	[0.48]	[-0.20]	[-2.04]	[-2.36]
Quintile	Cap	VW	0.18	0.17	0.05	-0.01	-0.19	-0.21
			[1.96]	[1.86]	[0.58]	[-0.07]	[-2.45]	[-2.71]
Decile	NYSE	VW	0.37	0.33	0.20	0.08	-0.03	-0.11
			[3.00]	[2.61]	[1.69]	[0.66]	[-0.28]	[-0.97]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.18	0.16	0.03		0.06	0.08
			[1.92]	[1.63]	[0.35]		[0.70]	[1.00]
Quintile	NYSE	EW	0.47	0.47	0.39	0.36	0.34	0.33
			[5.07]	[4.99]	[4.67]	[4.35]	[4.27]	[4.05]
Quintile	Name	VW	0.17	0.15	0.02		0.06	0.08
			[1.75]	[1.47]	[0.20]		[0.72]	[0.93]
Quintile	Cap	VW	0.15	0.14	0.03		0.09	0.11
			[1.61]	[1.47]	[0.35]		[1.18]	[1.37]
Decile	NYSE	VW	0.33	0.28	0.17	0.10		
			[2.63]	[2.19]	[1.40]	[0.82]		

Table 3: Conditional sort on size and CEB

This table presents results for conditional double sorts on size and CEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEB and short stocks with low CEB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		CEB Quintiles					CEB Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.42	0.82	0.90	0.98	0.92	0.50	0.47	0.43	0.34	0.42	0.36
		[1.74]	[3.46]	[3.86]	[4.16]	[3.38]	[4.55]	[4.26]	[3.89]	[3.11]	[3.90]	[3.30]
	(2)	0.66	0.71	0.85	0.87	0.92	0.26	0.29	0.20	0.17	0.12	0.10
		[2.80]	[3.13]	[3.98]	[4.03]	[4.00]	[2.74]	[2.97]	[2.12]	[1.72]	[1.30]	[1.08]
	(3)	0.47	0.79	0.78	0.84	0.95	0.48	0.48	0.39	0.36	0.31	0.29
		[2.24]	[3.82]	[3.80]	[4.15]	[4.53]	[5.15]	[5.11]	[4.30]	[3.91]	[3.30]	[3.09]
	(4)	0.58	0.69	0.76	0.74	0.83	0.25	0.22	0.11	0.12	-0.07	-0.04
		[2.98]	[3.50]	[3.96]	[3.92]	[4.18]	[2.53]	[2.20]	[1.19]	[1.22]	[-0.70]	[-0.47]
	(5)	0.52	0.61	0.58	0.51	0.59	0.07	0.07	-0.05	-0.10	-0.31	-0.33
		[3.11]	[3.54]	[3.55]	[3.07]	[3.62]	[0.68]	[0.66]	[-0.47]	[-1.01]	[-3.24]	[-3.42]

Panel B: Portfolio average number of firms and market capitalization

		CEB Quintiles					CEB Quintiles				
		Average n					Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	382	383	381	378	371	37	37	32	29	27
	(2)	108	108	108	108	107	56	56	57	55	55
	(3)	79	79	79	79	78	97	98	98	98	97
	(4)	66	67	67	67	66	204	211	214	211	206
	(5)	61	61	61	62	61	1727	1817	1671	1462	1375

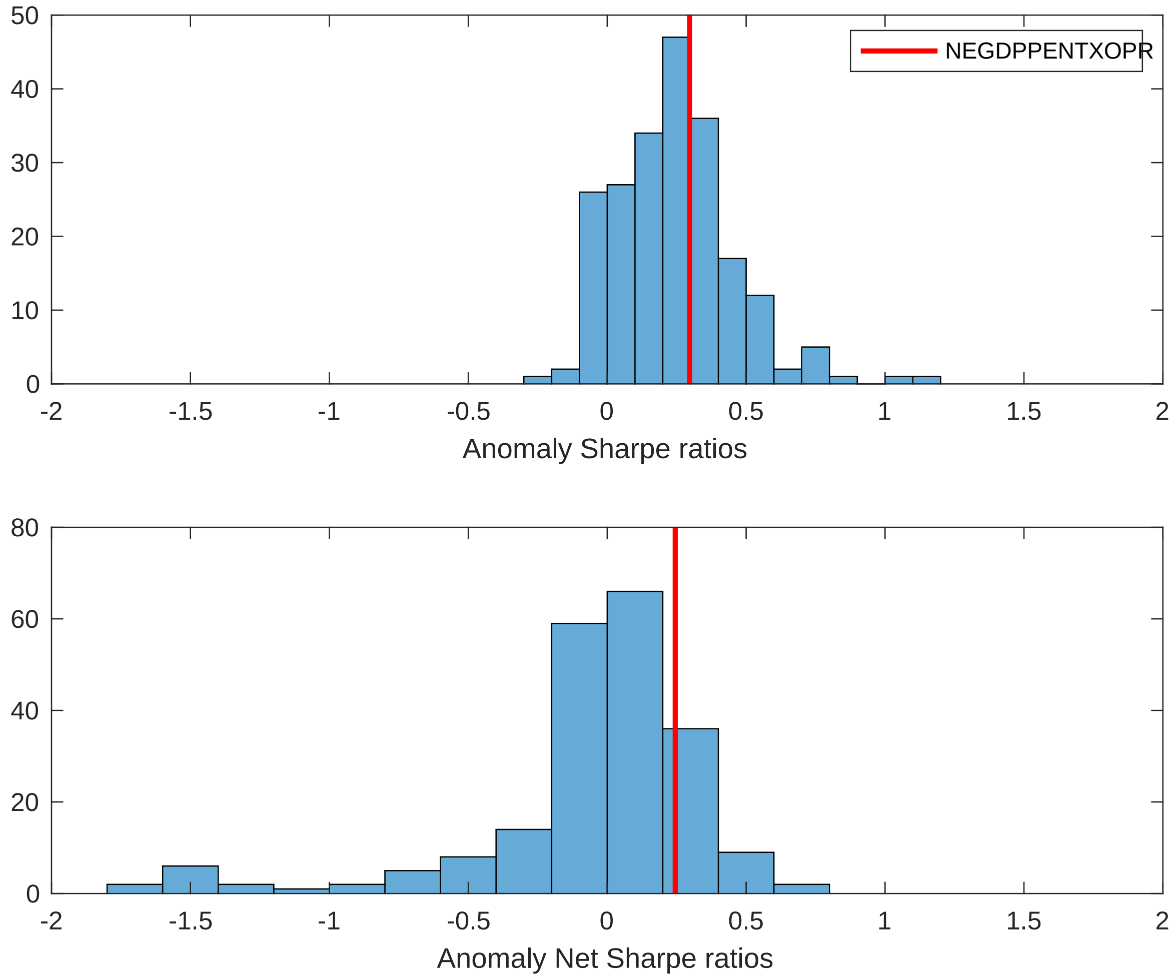


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

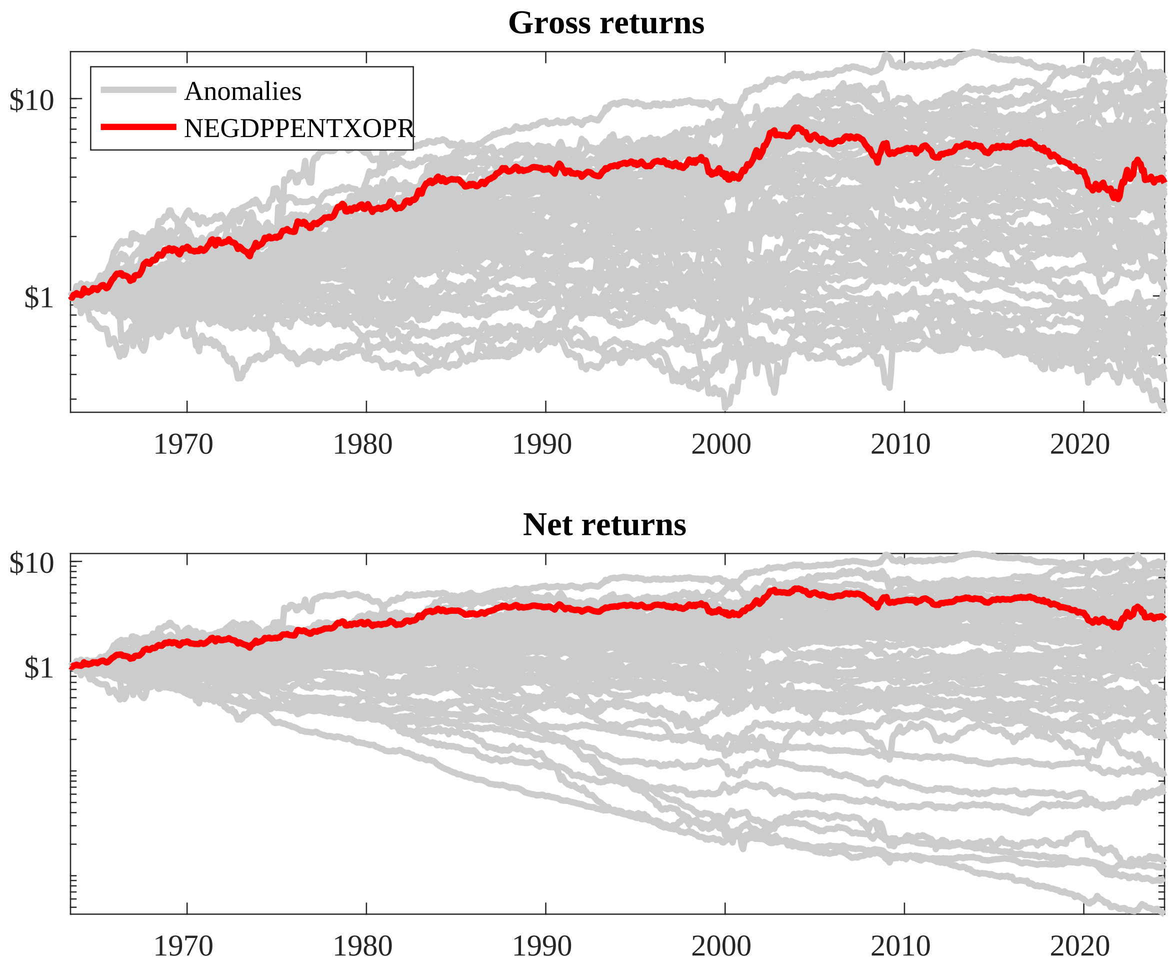
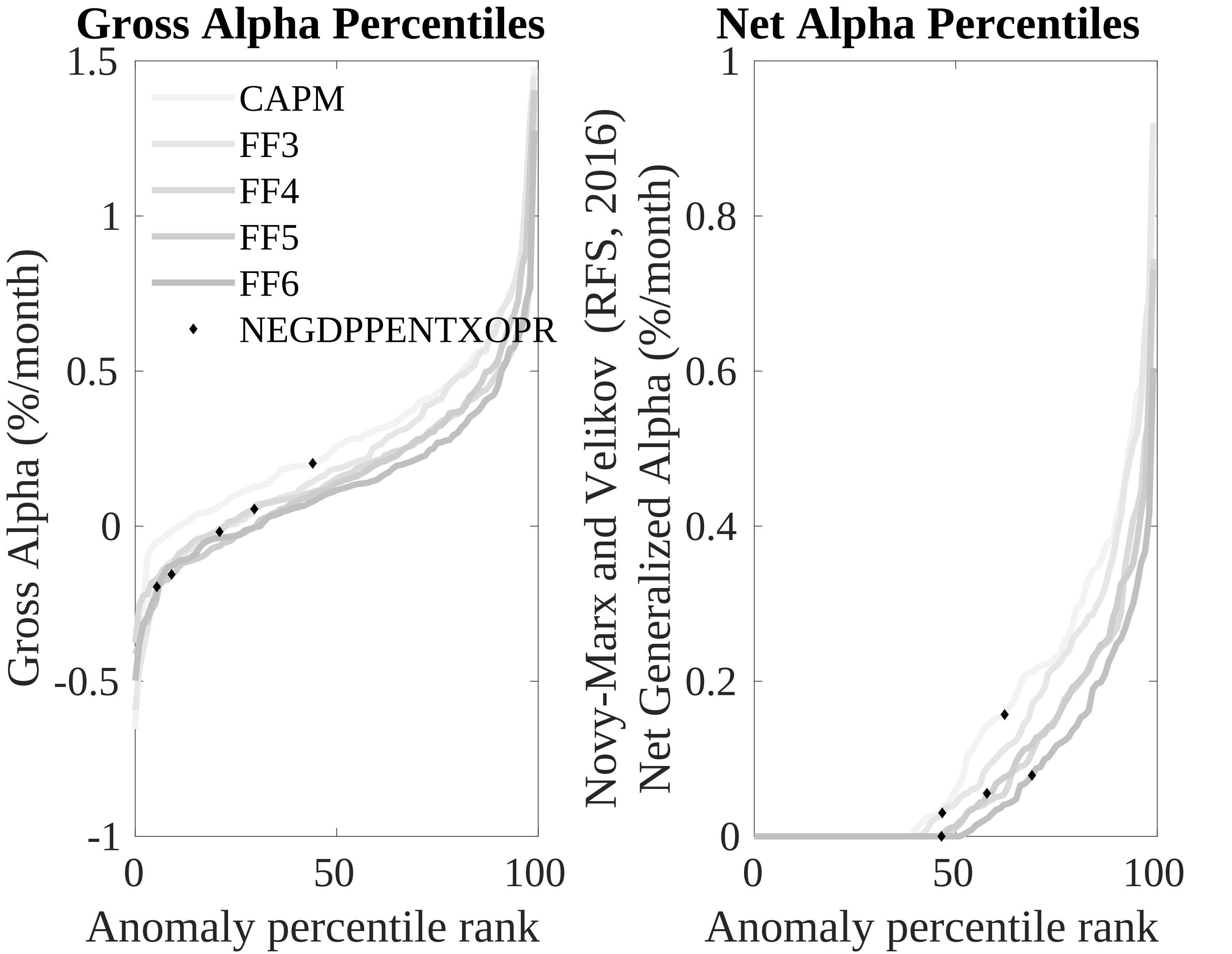
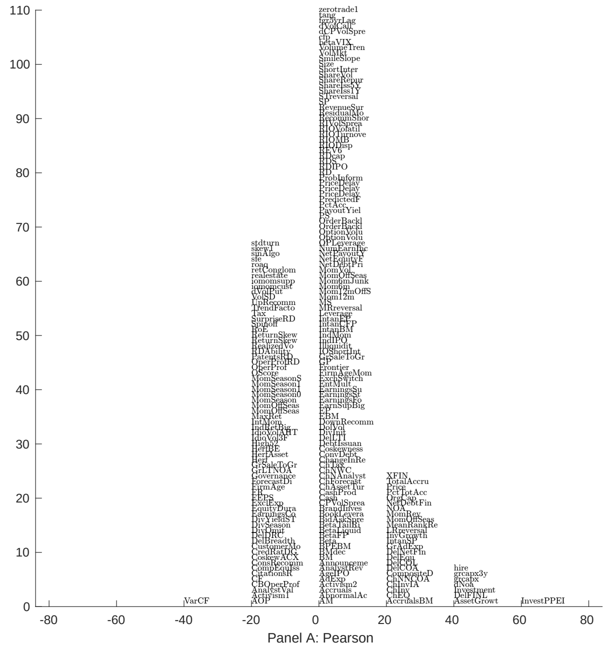


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDPPENTXOPR
Anomaly percentile rank
Anomaly percentile rank



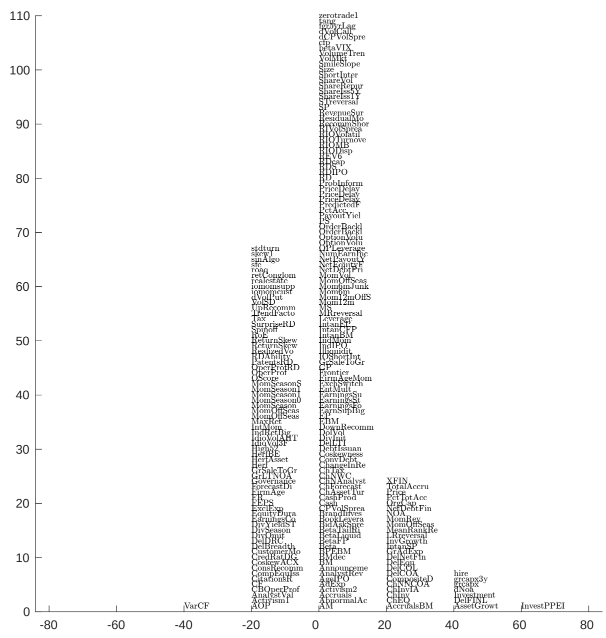


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

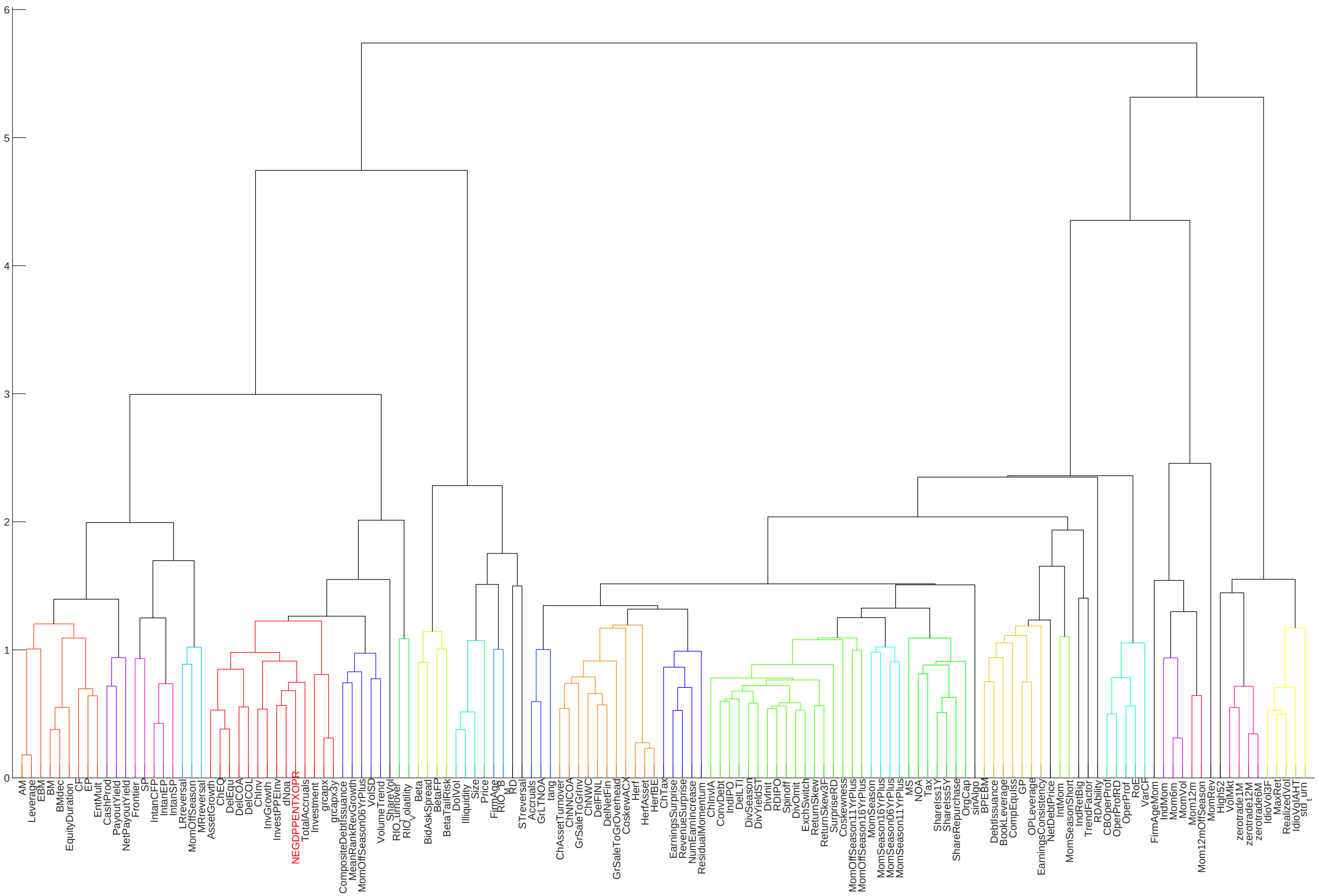


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

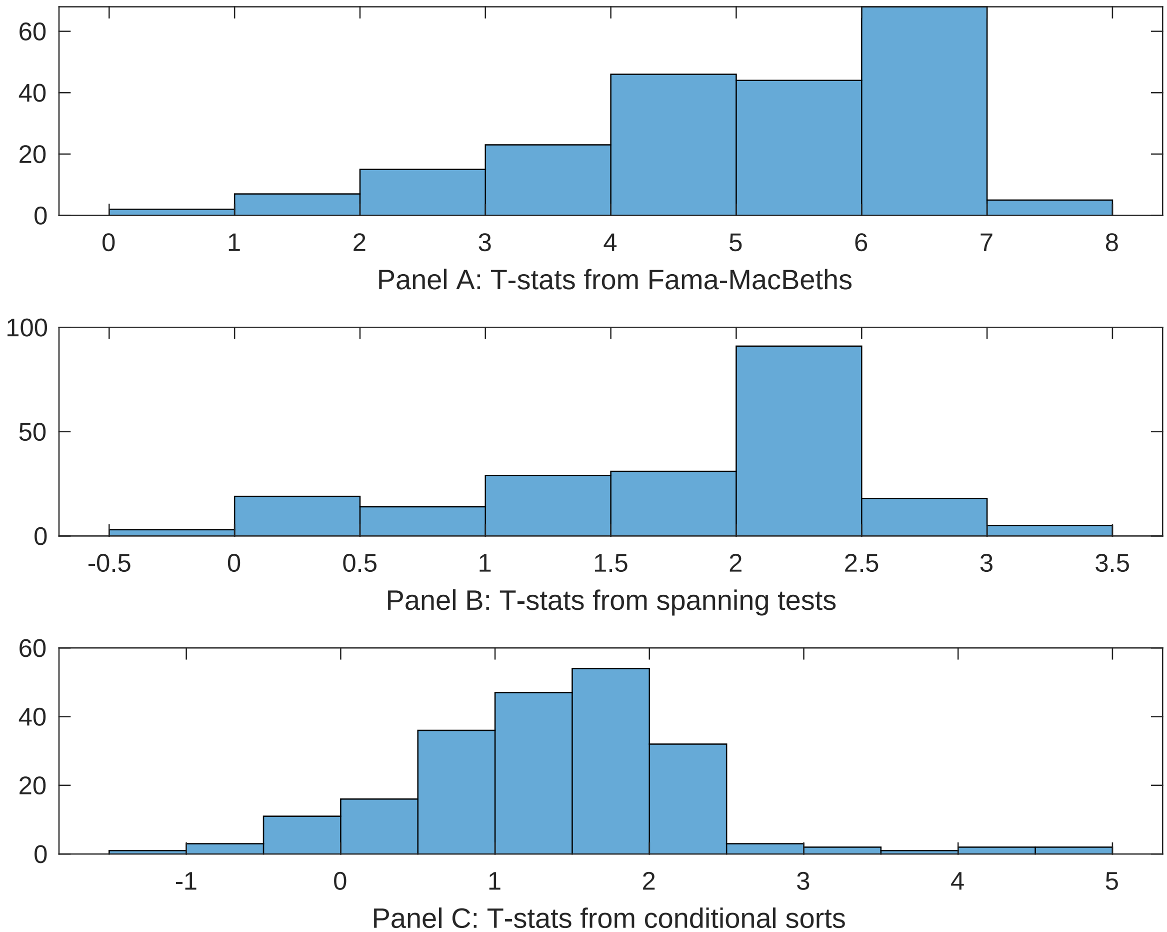


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEB} CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on CEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEB} CEB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.14	0.13	0.13	0.13	0.12	0.14
	[6.07]	[6.56]	[6.01]	[6.31]	[6.01]	[5.66]	[6.14]
CEB	0.26	0.35	0.41	0.29	0.33	0.45	0.25
	[2.72]	[3.30]	[4.02]	[3.15]	[3.43]	[5.08]	[2.24]
Anomaly 1	0.14						-0.16
	[7.51]						[-0.71]
Anomaly 2		0.10					0.34
		[7.25]					[1.54]
Anomaly 3			0.51				0.14
			[6.76]				[1.09]
Anomaly 4				0.91			0.47
				[7.94]			[2.83]
Anomaly 5					0.11		0.30
					[7.16]		[1.61]
Anomaly 6						0.75	0.15
						[5.44]	[1.11]
# months	732	727	727	732	720	713	708
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.18	-0.16	-0.18	-0.20	-0.25	-0.15	-0.21
	[-2.70]	[-2.19]	[-2.41]	[-2.50]	[-3.21]	[-1.87]	[-3.02]
Anomaly 1	50.72						39.87
	[15.77]						[11.06]
Anomaly 2		39.69					20.77
		[10.90]					[3.47]
Anomaly 3			36.63				4.08
			[9.92]				[0.68]
Anomaly 4				21.38			9.79
				[4.56]			[2.14]
Anomaly 5					33.58		9.10
					[7.80]		[2.06]
Anomaly 6						18.59	-5.82
						[4.29]	[-1.45]
mkt	11.68	11.56	11.84	12.33	12.47	12.35	10.99
	[7.11]	[6.53]	[6.61]	[6.58]	[6.82]	[6.55]	[6.90]
smb	26.83	21.13	24.44	27.14	29.21	29.80	21.99
	[11.29]	[7.96]	[9.30]	[9.91]	[11.06]	[10.91]	[8.95]
hml	0.91	-1.35	-0.41	5.78	3.84	2.59	-3.25
	[0.28]	[-0.39]	[-0.12]	[1.59]	[1.08]	[0.69]	[-1.02]
rmw	26.23	23.70	22.98	26.21	26.79	26.96	24.55
	[8.19]	[6.85]	[6.54]	[7.16]	[7.51]	[7.33]	[7.86]
cma	24.70	42.71	43.89	37.83	38.58	47.57	6.16
	[4.68]	[7.95]	[8.05]	[4.87]	[6.32]	[7.23]	[0.86]
umd	5.36	3.09	2.74	6.12	4.75	4.26	4.08
	[3.29]	[1.75]	[1.53]	[3.28]	[2.62]	[2.26]	[2.53]
# months	732	728	728	732	732	713	709
$\bar{R}^2$(%)	54	46	45	39	42	39	57

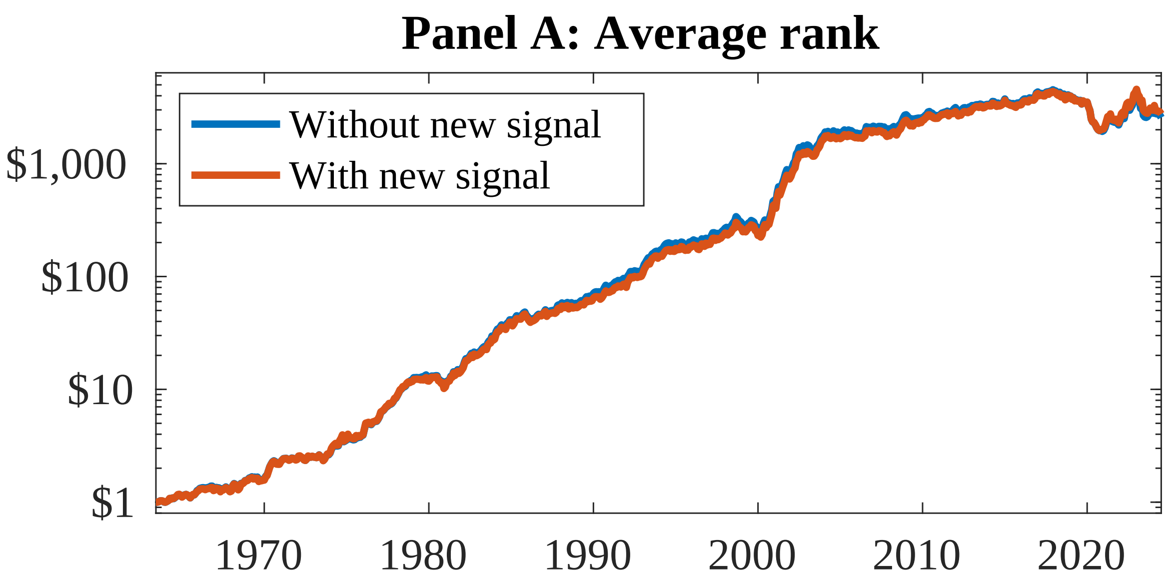


Figure 8: Combination strategy performance
This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004a). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Bansal, R. and Yaron, A. (2004b). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Breeden, D. T. (1979). An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics*, 7(3):265–296.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.

Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.

Hirshleifer, D., Hsu, P.-H., and Li, D. (2013). Innovative efficiency and stock returns. *Journal of Financial Economics*, 107(3):632–654.

Lucas, R. E. (1978). Asset prices in an exchange economy. *Econometrica*, 46(6):1429–1445.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper.*

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.